

Statistical Inference

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Parameter estimation

Let p be a pmf/pdf with parameters θ . We call $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$ a **sample** of p .

- How to estimate θ given the sample?
- How accurate is the estimate?

For example:

- Let $X_1, \dots, X_n$ be a sample of $\text{Poisson}(\lambda)$ with unknown parameter $\lambda > 0$. How to estimate λ using this sample?
- Let $X_1, \dots, X_n$ be a sample of $\mathcal{N}(\mu, \sigma^2)$ with unknown parameters μ and σ^2 , how to estimate μ and σ^2 using this sample?

We will first introduce two methods to estimate parameters. Then we consider the estimation errors.

Method of Moments

Definition. The k -th moment of a distribution is

$$\mu_k = \mathbb{E}(X^k)$$

Definition. The k -th central moment of a distribution is

$$\mu_k = \mathbb{E}\left((X - \mu)^k\right)$$

where $\mu = \mathbb{E}(X)$.

Remarks

- $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \mu_2 - \mu_1^2.$
- $\mu'_1 = \mathbb{E}(X - \mu) = \mathbb{E}(X) - \mu = 0.$
- $\mu'_2 = \mathbb{E}(X - \mu)^2 = \text{Var}(X).$

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$, where p is a pmf or pdf.

Definition. The k -th sample moment of a distribution is

$$m_k = \frac{1}{n} \sum_{i=1}^n X_i^k$$

Definition.

The k -th sample central moment of a distribution is

$$m'_k = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^k$$

where $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$.

Remarks

- The 1st sample moment m_1 is the sample mean, which is an unbiased estimator of μ :

$$\mathbb{E}(\bar{X}) = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = \frac{1}{n} \cdot (n\mu) = \mu$$

- $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of variance σ^2 .
- m_k is also an RV. We have its value only if we have realizations of $X_1, \dots, X_n$.
- $\mu'_2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ is also an estimator of σ^2 , but different from the unbiased estimator $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

Idea of the method of moments

- μ_k 's are functions of the parameter(s) θ .
- m_k 's are numbers when we have realizations of $X_1, \dots, X_n$.

Suppose we have k parameters in θ , we set a system of k equations of θ :

$$\begin{cases} \mu_1 = m_1 \\ \mu_2 = m_2 \\ \vdots \\ \mu_k = m_k \end{cases}$$

and solve for θ .

Example. Suppose we have $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$. Estimate λ using the method of moments.

Solution. We only have $k = 1$ parameter λ . Just need to solve one equation $\mu_1 = m_1$. As

$$\mu_1 = \mathbb{E}(X) = \lambda, \quad m_1 = \frac{1}{n} \sum_{i=1}^n X_i = \overline{X},$$

we have

$$\lambda = \overline{X}.$$

Hence $\overline{X}$ is the estimator of λ .

Example. Suppose we have $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Estimate μ and σ^2 using the method of moments.

Solution. Notice

$$\begin{aligned}\mu_1 &= \mu, & m_1 &= \overline{X}, \\ \mu_2 &= \mu^2 + \sigma^2, & m_2 &= \frac{1}{n} \sum_{i=1}^n X_i^2.\end{aligned}$$

Solving (μ, σ^2) from $\mu_1 = m_1$ and $\mu_2 = m_2$, we find the estimators

$$\mu = \overline{X}, \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \overline{X}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \overline{X})^2.$$

Method of Maximum Likelihood

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_\theta$, where p_θ is a pmf or pdf with parameter θ . Suppose $x_1, \dots, x_n$ are their realizations (observed values).

The joint distribution of the sample is

$$\prod_{i=1}^n p_\theta(x_i)$$

where x_i is the realization of X_i .

Definition. The likelihood function of θ is defined by

$$L(\theta) := \prod_{i=1}^n p_{\theta}(x_i).$$

Note that this is a function of θ as x_i 's are known.

Definition. We call the maximizer of $L(\theta)$:

$$\theta^* = \arg \max_{\theta} L(\theta)$$

the maximum likelihood estimator of θ .

Log-likelihood function

We usually take the (natural) logarithm of L to get the **log-likelihood function**:

$$l(\theta) := \ln L(\theta) = \sum_{i=1}^n \ln p_{\theta}(x_i).$$

Notice the logarithm is a monotonic increasing function, so l has the same maximizer as L .

It is almost always easier to maximize l than L .

Example. Given a sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$. Find the maximum likelihood estimator of λ .

Solution. Recall the pmf of $\text{Poisson}(\lambda)$ is $p_\lambda(x) = \frac{\lambda^x e^{-x}}{x!}$. The likelihood and log-likelihood functions are

$$L(\lambda) = \prod_{i=1}^n p_\lambda(x_i) = \prod_{i=1}^n \frac{\lambda^{x_i} e^{-x_i}}{x_i!},$$

$$l(\lambda) = \ln L(\lambda) = \sum_{i=1}^n (x_i \ln \lambda - \lambda - \ln(x_i!)).$$

Solution (cont)

To find the maximizer of $l(\lambda)$, we solve

$$l'(\lambda) = \sum_{i=1}^n \left(\frac{x_i}{\lambda} - 1 \right) = 0$$

for λ , and get the solution

$$\lambda = \frac{1}{n} \sum_{i=1}^n X_i = \overline{X}.$$

Hence $\overline{X}$ is the maximum likelihood estimator of λ .

Example

Given a sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Find the maximum likelihood estimator of μ and σ^2 .

Solution

Recall the pdf of $\mathcal{N}(\mu, \sigma^2)$ is $p_{\mu, \sigma^2}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{|x-\mu|^2}{2\sigma^2}}$. The likelihood and log-likelihood functions are

$$L(\mu, \sigma^2) = \prod_{i=1}^n p_{\mu, \sigma^2}(x_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{|x_i - \mu|^2}{2\sigma^2}},$$

$$l(\mu, \sigma^2) = \ln L(\lambda) = \sum_{i=1}^n \left(-\frac{(x_i - \mu)^2}{2\sigma^2} - \frac{1}{2} \ln(2\pi\sigma^2) \right).$$

Solution (cont)

To find the maximizer of $l(\mu, \sigma^2)$, we solve

$$\partial_{\mu} l(\mu, \sigma^2) = \sum_{i=1}^n \frac{x_i - \mu}{\sigma^2} = 0,$$

$$\partial_{\sigma^2} l(\mu, \sigma^2) = \sum_{i=1}^n \left(\frac{(x_i - \mu)^2}{2\sigma^4} - \frac{1}{2\sigma^2} \right) = 0$$

for μ, σ^2 , and get the solution

$$\mu = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}, \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Hence $\bar{X}$ and $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ are respectively the maximum likelihood estimators of μ and σ^2 .

Remarks

- There are mostly just one or two parameters in the distribution of interest.
- Method of moments (MoM) and maximum likelihood estimation (MLE) may give different answers.

Example

Given a sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Uniform}(0, b)$. Estimate b using the method of moments and the method of maximum likelihood.

Solution

We first consider the method of moments: since $\mu_1 = \frac{b}{2}$ and $m_1 = \bar{X}$, we get

$$b_{\text{MoM}} = 2\bar{X}.$$

Solution (cont)

Next consider the method of maximum likelihood: the likelihood function is

$$L(b) = \prod_{i=1}^n \frac{1}{b} = \frac{1}{b^n}, \quad \text{where } 0 \leq X_1, \dots, X_n \leq b.$$

In other words, we want to maximize $L(b)$ subject to $b \geq \max(X_1, \dots, X_n)$. Clearly, the maximizer is

$$b_{\text{MLE}} = \max(X_1, \dots, X_n).$$

We see b_{MoM} and b_{MLE} are different.

Estimation of standard errors

How accurate are the estimators (obtained by MoM and MLE)?

Idea:

Find the variance of the estimator, and substitute the unknowns by their estimates.

Example

Given a sample $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$. Find the standard error of the MoM estimate of λ .

Solution

Both mean and variance of $\text{Poisson}(\lambda)$ are λ . Recall the MoM estimator of λ is $\bar{X}$. Hence

$$\begin{aligned}\text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \cdot n\lambda = \frac{\lambda}{n} \approx \frac{\bar{X}}{n}.\end{aligned}$$

Another solution

Alternatively, we can use the unbiased estimator of $\text{Var}(X_i) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ and replace the second line above with

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \cdot n \text{Var}(X_i) = \frac{1}{n(n-1)} \sum_{i=1}^n (X_i - \bar{X})^2.$$

We see these two solutions are different, but they are both valid estimates of $\text{Var}(\bar{X})$.

Confidence Intervals

In statistics, instead of standard error of an estimator, it is more common to construct a **confidence interval (CI)** of the parameter to be estimated.

Definition. Let $\alpha \in (0, 1)$. An interval $[a, b]$ is called a $(1 - \alpha)100\%$ **confidence interval (CI)** of θ if

$$P(a \leq \theta \leq b) = 1 - \alpha.$$

- α is called the **significance level**. Typically α is set to 0.05, 0.01, or 0.1.
- $1 - \alpha$ is called the **confidence level**.

Note that $[a, b]$ is determined by α and $\hat{\theta}$.

Remarks

- Different value of $\hat{\theta}$ (due to different realizations of $X_1, \dots, X_n$) yields different $[a, b]$.
- Even the same $\hat{\theta}$ allows different choices of $[a, b]$. But we typically choose the one with smallest length.

A general method

If

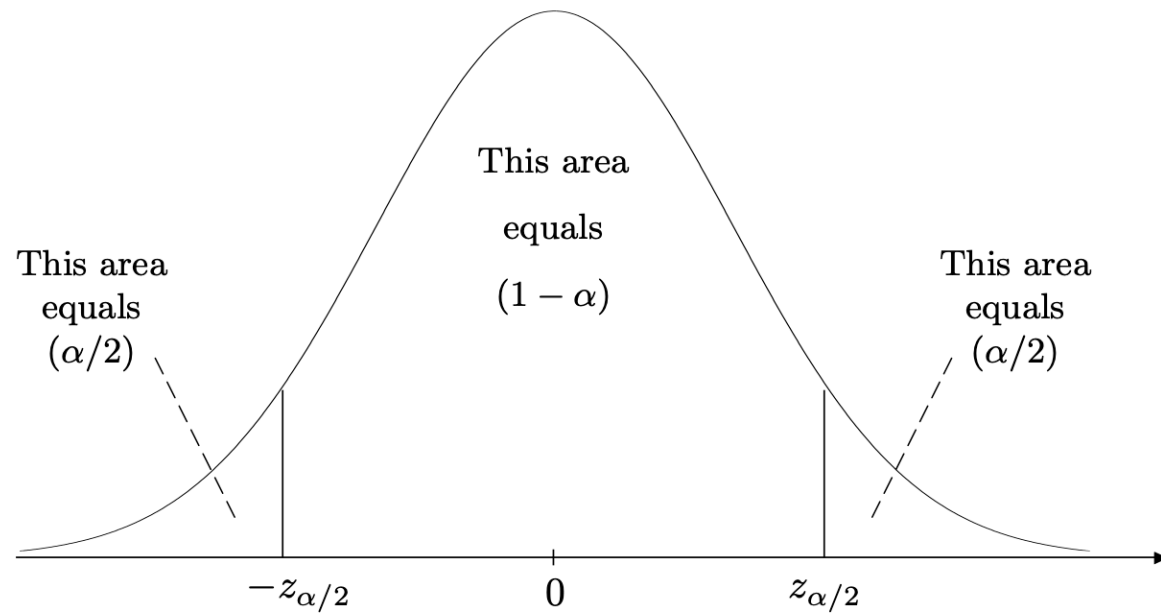
- $\mathbb{E}(\hat{\theta}) = \theta$,
- $\sigma(\hat{\theta})$ is known or can be accurately approximated,
- $\hat{\theta}$ follows Gaussian distribution,

then

$$Z = \frac{\hat{\theta} - \theta}{\sigma(\hat{\theta})} \sim \mathcal{N}(0, 1).$$

The pdf of $\mathcal{N}(0, 1)$:

Note that $z_{\alpha/2}$ is the value such that $\int_{z_{\alpha/2}}^{\infty} p(z)dz = \frac{\alpha}{2}$, where p is the pdf of $\mathcal{N}(0, 1)$.



Consequently, there is

$$P\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Substitute $Z = \frac{\hat{\theta} - \theta}{\sigma(\hat{\theta})}$ into above, we find

$$P\left(\hat{\theta} - z_{\alpha/2}\sigma(\hat{\theta}) \leq \theta \leq \hat{\theta} + z_{\alpha/2}\sigma(\hat{\theta})\right) = 1 - \alpha.$$

So the $(1 - \alpha)100\%$ CI of θ is

$$\left[\hat{\theta} - z_{\alpha/2}\sigma(\hat{\theta}), \quad \hat{\theta} + z_{\alpha/2}\sigma(\hat{\theta})\right].$$

Remarks

The CI is often written as

$$\hat{\theta} \pm z_{\alpha/2} \sigma(\hat{\theta})$$

- $\hat{\theta}$ is called the **center**.
- $z_{\alpha/2} \sigma(\hat{\theta})$ is called the **margin**.

One- and Two-sample Cases

1. One-sample case

- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$ with mean μ and variance σ^2 . Sample size n .

2. Two-sample case

- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_X$ with mean μ_X and variance σ_X^2 . Sample size n .
- $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} p_Y$ with mean μ_Y and variance σ_Y^2 . Sample size m .
- These two samples are independent.

Cases of CIs to be considered:

Case	1	2
A	CI of μ . Any p , known σ^2 .	CI of $\mu_X - \mu_Y$. Any p_X, p_Y , known σ_X^2, σ_Y^2 .
B	CI of μ . Any p , unknown σ^2 , large n .	CI of $\mu_X - \mu_Y$. Any p_X, p_Y , unknown σ_X^2, σ_Y^2 , large n, m .
C	CI of μ . Gaussian p , unknown σ^2 , small n .	CI of $\mu_X - \mu_Y$. Gaussian p_X, p_Y , unknown σ_X^2, σ_Y^2 , small n, m .
D	CI of σ^2 . Gaussian p .	CI of σ_X^2 / σ_Y^2 . Gaussian p_X, p_Y .

Case A1: CI of μ for any distribution p with known σ^2

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$ with unknown mean μ but known σ^2 . Now we construct the CI of μ .

We choose $\hat{\mu} = \bar{X}$ as the estimator of μ . Note that

- $\mathbb{E}(\bar{X}) = \mu$.
- We know $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$ if
 - p is Gaussian, or
 - p is not Gaussian, but n is large (e.g., $n \geq 30$) by Central Limit Theorem (CLT).

As $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1)$, there is

$$P\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Namely

$$P\left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

So the $(1 - \alpha)100\%$ CI of μ is

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}.$$

Example

Suppose we have a sample

2.5, 8.4, 8.0, 4.5, 7.4, 9.2

from a normal distribution with $\sigma = 2.2$. Find the 95% CI of the mean μ .

Solution

The confidence level is $\alpha = 0.05$ and hence $z_{\alpha/2} = z_{0.025} = 1.96$. The sample mean is $\bar{X} = 6.5$ and sample size is $n = 6$. Hence the 95% CI of μ is

$$\bar{X} \pm z_{0.025} \frac{\sigma}{\sqrt{n}} = 6.5 \pm 1.96 \cdot \frac{2.2}{\sqrt{6}}.$$

A side question: Sample size requirement?

In the case above, how large n needs to be if we want sufficiently small margin Δ ?

In other words, we need

$$z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \Delta.$$

Solving for n , we find

$$n \geq \left(\frac{z_{\alpha/2} \sigma}{\Delta} \right)^2.$$

Round n up if the right-hand-side is non-integer.

Case A2: CI of $\mu_X - \mu_Y$ for any distributions p_X, p_Y with known variances σ_X^2, σ_Y^2

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_X$ with unknown mean μ_X but known variance σ_X^2 , and independently $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} p_Y$ with unknown mean μ_Y but known variance σ_Y^2 . Now we construct the CI of $\mu_X - \mu_Y$.

We choose $\bar{X}$ and $\bar{Y}$ as the estimators of μ_X and μ_Y , respectively. Note that

- $\mathbb{E}(\bar{X}) = \mu_X$ and $\mathbb{E}(\bar{Y}) = \mu_Y$.
- We know $\bar{X} \sim \mathcal{N}(\mu_X, \frac{\sigma_X^2}{n})$ and $\bar{Y} \sim \mathcal{N}(\mu_Y, \frac{\sigma_Y^2}{m})$ if
 - p_X and p_Y are Gaussian, or
 - p_X (or p_Y) is not Gaussian, but n (or m) is large.

Hence $\bar{X} - \bar{Y} \sim \mathcal{N}\left(\mu_X - \mu_Y, \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}\right)$ and there is

$$P\left(-z_{\alpha/2} \leq \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Namely

$$P\left((\bar{X} - \bar{Y}) - z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}} \leq \mu_X - \mu_Y \leq (\bar{X} - \bar{Y}) + z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}\right) = 1 - \alpha.$$

So the $(1 - \alpha)100\%$ CI of $(\mu_X - \mu_Y)$ is

$$(\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}}.$$

Case B1: CI of μ for any p with unknown σ^2 but large sample size n

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$ with unknown mean μ and variance σ^2 but large n .

Note that

- $\mathbb{E}(\bar{X}) = \mu$.
- $\bar{X} \sim \mathcal{N}(\mu, \frac{\sigma^2}{n})$ by CLT.

We can approximate σ^2 by $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ (or $\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$), and find the $(1 - \alpha)100\%$ CI of μ as

$$\bar{X} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}.$$

Example. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Binomial}(p)$ where $n \geq 30$. Find the $(1 - \alpha)100\%$ CI of p .

Solution. Recall the mean and variance of $\text{Binomial}(p)$ are p and $p(1 - p)$, respectively.

We know $\mathbb{E}(\bar{X}) = p$ and by CLT that $\bar{X} \sim \mathcal{N}(p, \frac{p(1-p)}{n})$. Furthermore, we approximate the variance $p(1 - p)$ by

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \dots = \bar{X}(1 - \bar{X}). \quad (\text{Hint: } X_i \text{ is 1 or 0, so } X_i^2 = X_i).$$

Then the $(1 - \alpha)100\%$ CI of p is

$$\bar{X} \pm z_{\alpha/2} \sqrt{\frac{\bar{X}(1 - \bar{X})}{n}}.$$

Case B2: CI of $\mu_X - \mu_Y$ for any p_X, p_Y with unknown σ_X^2, σ_Y^2 but large n, m

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_X$ with unknown mean μ_X and variance σ_X^2 , and independently $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} p_Y$ with unknown mean μ_Y and variance σ_Y^2 . Now we construct the CI of $\mu_X - \mu_Y$.

We choose $\bar{X}$ and $\bar{Y}$ as the estimators of μ_X and μ_Y , respectively. Note that

- $\mathbb{E}(\bar{X}) = \mu_X$ and $\mathbb{E}(\bar{Y}) = \mu_Y$.
- We know $\bar{X} \sim \mathcal{N}(\mu_X, \frac{\sigma_X^2}{n})$ and $\bar{Y} \sim \mathcal{N}(\mu_Y, \frac{\sigma_Y^2}{m})$ by CLT.

We can approximate σ_X^2 and σ_Y^2 by their sample variances, and construct the CI of $\mu_X - \mu_Y$ similarly.

Example. Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Binomial}(p_X)$ and $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} \text{Binomial}(p_Y)$ independently, where $n, m \geq 30$. Find the $(1 - \alpha)100\%$ CI of $p_X - p_Y$.

Solution. We know $\mathbb{E}(\bar{X}) = p_X$ and $\mathbb{E}(\bar{Y}) = p_Y$, and by CLT that

$$\bar{X} \sim \mathcal{N}\left(p_X, \frac{p_X(1 - p_X)}{n}\right) \quad \text{and} \quad \bar{Y} \sim \mathcal{N}\left(p_Y, \frac{p_Y(1 - p_Y)}{m}\right)$$

We approximate the variances $p_X(1 - p_X)$ by

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \dots = \bar{X}(1 - \bar{X}).$$

Similar for $p_Y(1 - p_Y)$. Then the $(1 - \alpha)100\%$ CI of $(p_X - p_Y)$ is

$$(\bar{X} - \bar{Y}) \pm z_{\alpha/2} \sqrt{\frac{\bar{X}(1 - \bar{X})}{n} + \frac{\bar{Y}(1 - \bar{Y})}{m}}.$$

Case C1: CI of μ for Gaussian p with unknown σ^2 and small sample size n

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and σ^2 and small n (e.g., $n < 30$).

We can approximate σ^2 by its unbiased estimator

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

but n is small so s^2 is not accurate.

In such case, $\frac{\bar{X} - \mu}{s/\sqrt{n}}$ no longer follows $\mathcal{N}(0, 1)$!

Instead, $\frac{\bar{X} - \mu}{s/\sqrt{n}}$ follows the Student's t distribution with degrees of freedom $n - 1$.

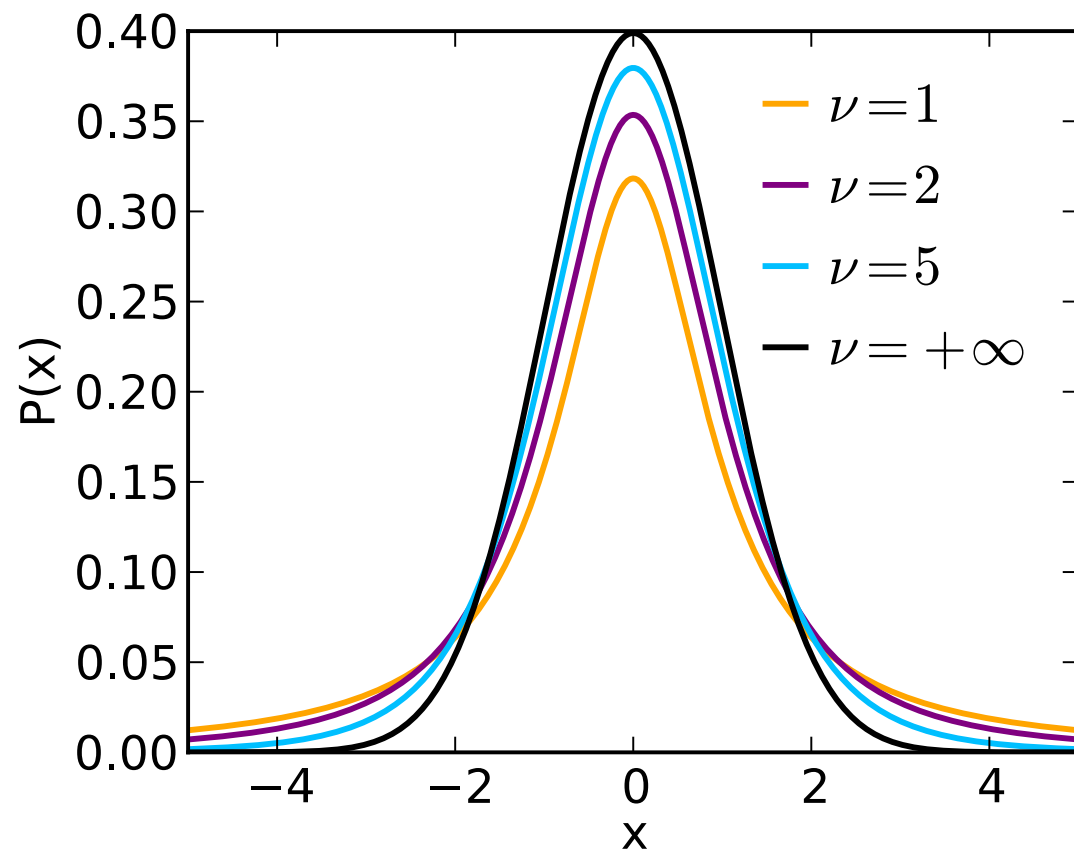
Student's t distribution

Definition. Suppose $Z \sim \mathcal{N}(0, 1)$ and $V \sim \chi^2_\nu$ are independent, then

$$t = \frac{Z}{\sqrt{V/\nu}}$$

is said to follow the **Student's t distribution with degrees of freedom ν** , denoted by $t \sim t_\nu$.

The pdf of Student's t_ν distribution with degrees of freedom ν



Student's t distribution

The pdf of t_ν is (no need to memorize)

$$p(t) = \frac{1}{\sqrt{\nu} B(\frac{1}{2}, \frac{1}{\nu})} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}$$

for every $t \in \mathbb{R}$, where B is the pdf of Beta distribution.

- p is symmetric about $t = 0$.
- As $\nu \rightarrow \infty$, $p \rightarrow \mathcal{N}(0, 1)$.

Proposition

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with unknown μ and σ^2 , and n is small. Denote

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Then

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}.$$

Sketch of proof

We rewrite

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\frac{s/\sqrt{n}}{\sigma/\sqrt{n}}} = \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\frac{s}{\sigma}} = \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)s^2}{\sigma^2} / (n-1)}}.$$

We just need show

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1), \quad \frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

and these two are independent RVs.

By the proposition, we have

$$P\left(-t_{\alpha/2, n-1} \leq \frac{\bar{X} - \mu}{s/\sqrt{n}} \leq t_{\alpha/2, n-1}\right) = 1 - \alpha,$$

or equivalently

$$P\left(\bar{X} - t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}\right) = 1 - \alpha.$$

Hence the $(1 - \alpha)100\%$ CI of μ is

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

Example

A Gaussian sample of size $n = 18$ has $\bar{X} = 0.29$ and $s^2 = 0.074^2$. Find the 95% CI of its mean μ .

Solution

Notice that $\alpha = 0.05$ and

$$t_{\alpha/2, n-1} = t_{0.025, 17} = 2.898.$$

Hence the 95% CI of μ is

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} = 0.29 \pm 2.898 \cdot \frac{0.074}{\sqrt{18}} = 0.29 \pm 0.05.$$

Case C2: CI of $\mu_X - \mu_Y$ from two Gaussian p_X, p_Y with unknown variances σ_X^2, σ_Y^2 and small sample sizes n, m

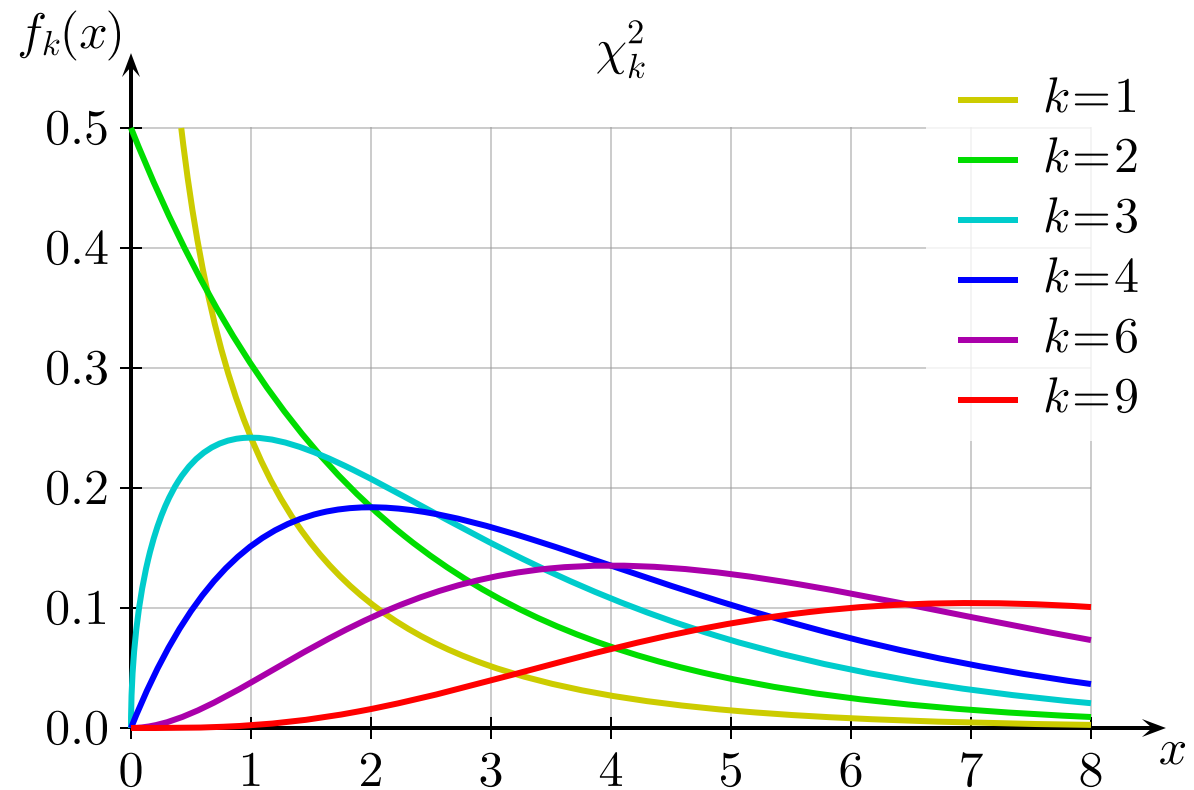
Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p_X = \mathcal{N}(\mu_X, \sigma_X^2)$ with unknown mean μ_X and variance σ_X^2 , and independently $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} p_Y = \mathcal{N}(\mu_Y, \sigma_Y^2)$ with unknown mean μ_Y and variance σ_Y^2 . Now we construct the CI of $\mu_X - \mu_Y$.

We choose $\bar{X}$ and $\bar{Y}$ as the estimators of μ_X and μ_Y , and s_X^2 and s_Y^2 as the unbiased estimators of σ_X^2 and σ_Y^2 .

We will consider two subcases:

- Subcase C2(i): $\sigma_X^2 = \sigma_Y^2$.
- Subcase C2(ii): $\sigma_X^2 \neq \sigma_Y^2$.

The pdf of χ_k^2 with degrees of freedom k



Subcase C2(i): CI of $\mu_X - \mu_Y$ with $\sigma_X^2 = \sigma_Y^2 = \sigma^2$

Notice that

$$\frac{(n-1)s_X^2}{\sigma_X^2} + \frac{(m-1)s_Y^2}{\sigma_Y^2} = \frac{(n-1)s_X^2 + (m-1)s_Y^2}{\sigma^2} \sim \chi_{n+m-2}^2.$$

We define the **pooled sample variance** by

$$s_p^2 = \frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{j=1}^m (Y_j - \bar{Y})^2}{n+m-2}.$$

Then it is shown that the $(1 - \alpha)100\%$ CI of $(\mu_X - \mu_Y)$ is

$$(\bar{X} - \bar{Y}) \pm t_{\frac{\alpha}{2}, n+m-2} s_p \sqrt{\frac{1}{n} + \frac{1}{m}}.$$

Subcase C2(ii): CI of $\mu_X - \mu_Y$ with $\sigma_X^2 \neq \sigma_Y^2$

Determine the degrees of freedom ν (round it to the nearest integer) by

$$\nu = \frac{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}{\frac{s_X^4}{n^2(n-1)} + \frac{s_Y^4}{m^2(m-1)}}.$$

Then it is shown that the $(1 - \alpha)100\%$ CI of $(\mu_X - \mu_Y)$ is

$$(\bar{X} - \bar{Y}) \pm t_{\frac{\alpha}{2}, \nu} \sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}.$$

Case D1: CI of σ^2 in $\mathcal{N}(\mu, \sigma^2)$

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with sample mean $\bar{X}$ and unbiased sample variance s^2 .

It is known that

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Therefore

$$P\left(\chi_{1-\frac{\alpha}{2}, n-1} \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha.$$

Hence the $(1 - \alpha)100\%$ CI of σ^2 is

$$\left[\frac{(n-1)s^2}{\chi_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)s^2}{\chi_{1-\frac{\alpha}{2}, n-1}} \right].$$

Case D2: CI of σ_X^2/σ_Y^2 for two Gaussian $\mathcal{N}(\mu_X, \sigma_X^2)$ and $\mathcal{N}(\mu_Y, \sigma_Y^2)$

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_X, \sigma_X^2)$ with unbiased sample variance s_X^2 , and independently $Y_1, \dots, Y_m \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_Y, \sigma_Y^2)$ with unbiased sample variance s_Y^2 .

We know

$$\frac{(n-1)s_X^2}{\sigma_X^2} \sim \chi_{n-1}^2, \quad \frac{(m-1)s_Y^2}{\sigma_Y^2} \sim \chi_{m-1}^2.$$

We need a new distribution, called the F -distribution, to find the CI of σ_X^2/σ_Y^2 .

F -distribution

Definition

Suppose $U \sim \chi_n^2$ and $V \sim \chi_m^2$ are independent, then the ratio

$$F = \frac{U/n}{V/m}$$

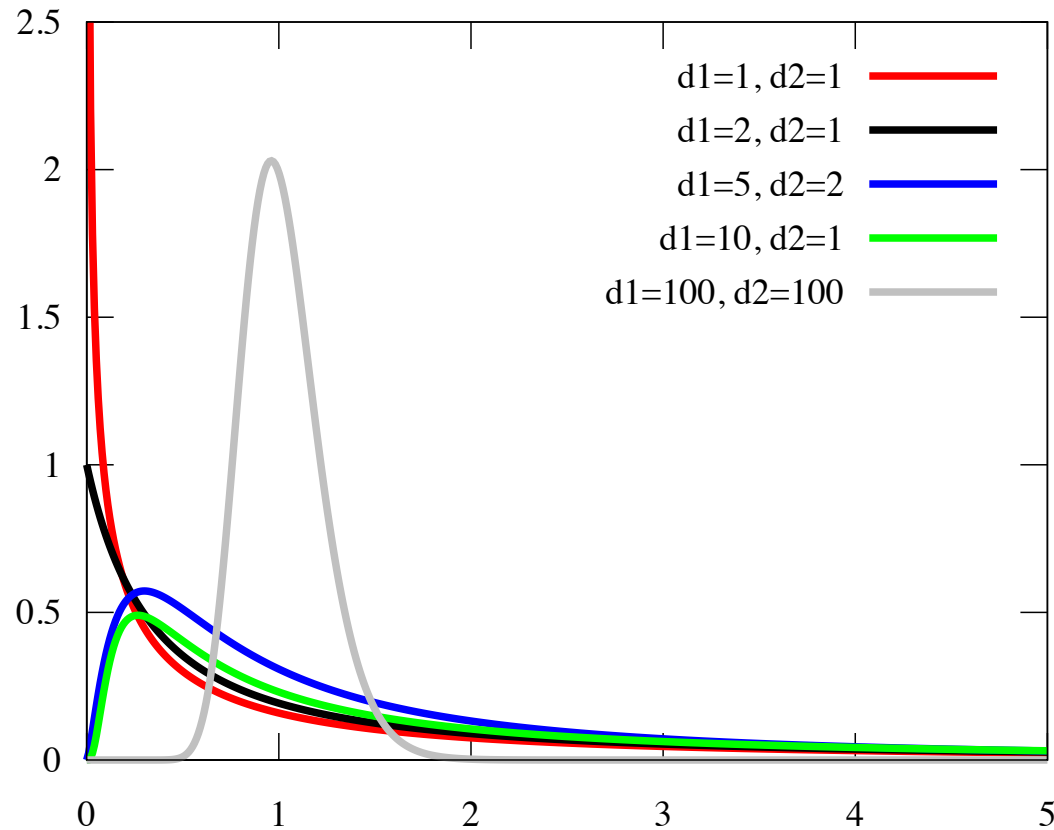
is said to follow the F -distribution with numerator degrees of freedom n and denominator degrees of freedom m . This is denoted by $F \sim F_{n,m}$.

Remarks

- $F = \frac{U/n}{V/m}$ is nonnegative.
- If $F = \frac{U/n}{V/m} \sim F_{n,m}$, then $\frac{1}{F} = \frac{V/m}{U/n} \sim F_{m,n}$.

The pdf of F_{d_1, d_2} distribution with degrees of freedom d_1, d_2

- d_1 (d_2) is numerator (denominator) degrees of freedom.



Now we know

$$\frac{s_X^2/\sigma_X^2}{s_Y^2/\sigma_Y^2} \sim F_{n-1,m-1}.$$

Therefore

$$P\left(F_{1-\frac{\alpha}{2},n-1,m-1} \leq \frac{s_X^2/\sigma_X^2}{s_Y^2/\sigma_Y^2} \leq F_{\frac{\alpha}{2},n-1,m-1}\right) = 1 - \alpha.$$

Hence the $(1 - \alpha)100\%$ CI of σ_X^2/σ_Y^2 is

$$\left[\frac{s_X^2/s_Y^2}{F_{\frac{\alpha}{2},n-1,m-1}}, \frac{s_X^2/s_Y^2}{F_{1-\frac{\alpha}{2},n-1,m-1}} \right].$$

Hypothesis Testing

A dual concept of CI is **Hypothesis Testing**: suppose someone presumes the parameter θ of interest equals θ_0 , we want to see whether this hypothesis should be rejected based on sample.

Formally, this hypothesis test is represented as

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_A : \theta \neq \theta_0.$$

Here H_0 is called the **null hypothesis**, and H_A is called the **alternative hypothesis**.

The test above is called **two-sided**. There are also **one-sided** tests:

- If $H_A : \theta < \theta_0$, then the test is called **left-tailed**.
- If $H_A : \theta > \theta_0$, then the test is called **right-tailed**.

Goal of Hypothesis Testing

The goal is to decide whether H_0 should be rejected based on the estimate of θ .

Specifically, consider the two-sided hypothesis test above. To decide whether we should reject H_0 , we need to specify a **significance level** α . Then we find the estimate $\hat{\theta}$ of θ based on a sample, and reject H_0 if it holds with probability less than α given this observation $\hat{\theta}$.

Example

A Gaussian sample of size $n = 18$ has sample mean $\bar{X} = 0.29$ and unbiased sample variance $s^2 = 0.074^2$. Consider the hypothesis test

$$H_0 : \mu = 0.2 \quad \text{vs} \quad H_A : \mu \neq 0.2$$

with significance level $\alpha = 0.05$.

Solution

We know $t = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$. Under $H_0 : \mu = 0.2$, there is $t = \frac{0.29 - 0.2}{0.074/\sqrt{18}} = 5.16$.

On the other hand, $t_{\alpha/2, n-1} = t_{0.025, 17} = 2.11$. We call $[-2.11, 2.11]$ the **acceptance region** of H_0 , and $(-\infty, -2.11) \cup (2.11, \infty)$ the **rejection region** of H_0 .

Since $t > 2.11$ and is in the rejection region, H_0 should be rejected.

Example

Two Gaussian distributions with unequal variances each having sample:

$$\begin{aligned} n &= 12, & \bar{X} &= 4.8, & s_X^2 &= 1.6^2 \\ m &= 18, & \bar{Y} &= 5.3, & s_Y^2 &= 1.4^2 \end{aligned}$$

Consider the hypothesis test

$$H_0 : \mu_X = \mu_Y \quad \text{vs} \quad H_A : \mu_X < \mu_Y$$

with significance level $\alpha = 0.05$.

Solution

Since their variances are equal, we can compute the pooled sample variance

$$s_p^2 = \frac{(n-1)s_X^2 + (m-1)s_Y^2}{n+m-2} = \frac{(12-1) \cdot 1.6^2 + (18-1) \cdot 1.4^2}{12+18-2} = 1.4818^2.$$

Under $H_0 : \mu_X = \mu_Y$, there is

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{s_p \sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t_{n+m-2}$$

Plugging in numbers, we find $t = -0.9054$. On the other hand,

$t_{\alpha/2, n+m-2} = t_{0.025, 28} = 1.654$. Note the rejection region of H_0 is $(-\infty, -1.654)$. Since $t > -1.654$, we should not reject H_0 .

Example

Two Gaussian distributions with unequal variances each having sample:

$$\begin{aligned} n &= 30, & \bar{X} &= 6.7, & s_X^2 &= 0.6^2 \\ m &= 20, & \bar{Y} &= 7.5, & s_Y^2 &= 1.2^2 \end{aligned}$$

Consider the hypothesis test

$$H_0 : \mu_X - \mu_Y = -0.2 \quad \text{vs} \quad H_A : \mu_X - \mu_Y < -0.2$$

with significance level $\alpha = 0.05$.

Solution

Since their variances are not equal, we know

$$t = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}} \sim t_\nu$$

where the degrees of freedom ν is the integer nearest to

$$\frac{\frac{s_X^2}{n} + \frac{s_Y^2}{m}}{\frac{s_X^4}{n^2(n-1)} + \frac{s_Y^4}{m^2(m-1)}}.$$

Solution (cont)

Plugging in the numbers, we find the fraction above is 25.4 (so we take $\nu = 25$), and

$$t = \frac{6.7 - 7.5 - (-0.2)}{\sqrt{\frac{0.6^2}{30} + \frac{1.2^2}{20}}} = -2.07$$

under $H_0 : \mu_X - \mu_Y = -0.2$.

On the other hand, $t_{\alpha, \nu} = t_{0.05, 25} = 1.708$. Note the acceptance region of H_0 is $[-1.708, \infty)$.

Since $t < -1.708$, we should reject H_0 .

Type I and Type II errors

	H_0 is true	H_0 is false
H_0 is not rejected	correct (with probability $1 - \alpha$)	Type II error (with probability β)
H_0 is rejected	Type I error (with probability α)	correct (with probability $1 - \beta$)

- Type I error is also called "false positive". Its probability is α when the test has a prescribed significance level α .
- Type II error is also called "false negative". Its probability is denoted by β . Note $1 - \beta$ is known as **the power of the test**, which is the larger the better.
- Type I error is more notorious than Type II error. In practice, people usually prescribe a small α (e.g., 0.05) and try to minimize β when designing a hypothesis test.

p-value

The *p*-value is the probability of obtaining a test statistic at least as extreme as the one obtained under H_0 .

Example

Suppose $Z \sim \mathcal{N}(\mu, 1)$ and we observe a value 3.61 of Z . Consider $H_0 : \mu = 0$ vs $H_A : \mu > 0$ (Note this is a one-sided right-tailed test). Then the *p*-value is

$$P(Z > 3.61) = 1 - \Phi(3.61) = 0.0002,$$

where Φ is the cdf of $\mathcal{N}(0, 1)$.

This *p*-value is well below 0.05, so we should reject H_0 if the significance level is set to $\alpha = 0.05$.

Example

Suppose $Z \sim \mathcal{N}(\mu, 1)$ and we observe a value -0.94 of Z . Consider $H_0 : \mu = 0$ vs $H_A : \mu \neq 0$ (Note this is a two-sided test). Then the p -value is

$$P(Z > 0.94 \text{ or } Z < -0.94) = 2(1 - \Phi(0.94)) = 0.3472.$$

This p -value is quite large, so we should not reject H_0 if the significance level is set to $\alpha = 0.05$.